

WORK PROCEDURE

PRO-MEC-010A

WORK PROCEDURE FOR CHANGE, MAINTENANCE OR REPARATION ON THE CUBEX ACTUATOR

OBJECTIVES

The objective of this procedure is to provide a safe way to change, repair or do maintenance to an actuator on a drill. The procedure covers work in the workshop, on a drill site (stope) and in the underground workshop not equipped with an overhead crane.

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL. It applies in the following cases:

- **For a change, cleaning or maintenance of the actuator**
 - **In a stope**
 - **In the shop with a lifting tool (overhead crane, lifting hand)**

INTERPRETATION

The personnel having the authority for the interpretation of this document are the Maintenance Manager or the Operations Manager. Any request for modification, revision or update must be made to one of these persons.

RESPONSABILITIES

The worker is responsible for using this procedure. The supervisor has the responsibility to enforce the procedure and that the worker has the skills to perform the work. It must also provide the resources for the work.

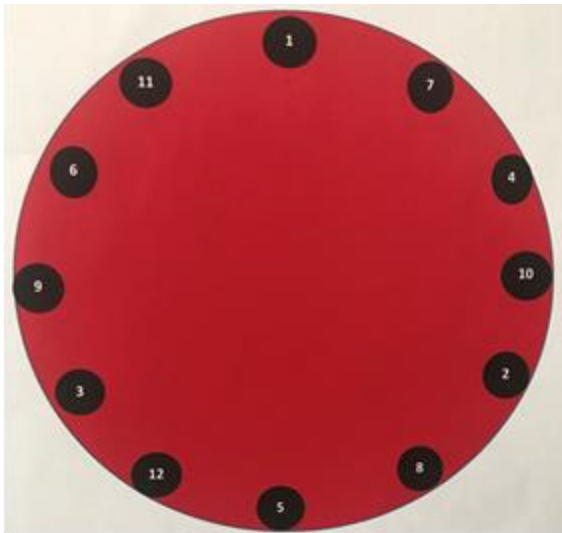
PROCEDURE FOR CHANGING, CLEANING OR PERFORMING MAINTENANCE ON THE ACTUATOR IN THE SHOP EQUIPPED WITH A CRANE OR LIFTING HAND

The supervisor will designate a team leader who will take charge of all operations. This team leader must be a supervisor or a mechanic and will have to meet the operator of the lifting equipment to explain the procedure in detail. In addition, he will be the only one authorized to communicate with the operator of the lifting equipment during the operation.

1. Lifting equipment must have a capacity of 6,000 pounds or more;
2. Position the drill level so that it is stable;

WORK PROCEDURE FOR CHANGE, MAINTENANCE OR REPARATION ON THE CUBEX ACTUATOR

3. In the case of a lifting hand, position the drill so that the mast is centered with the lifting point and thus eliminate the risk of swinging when the mast is lifted and detached from the actuator;
4. Position the gear box five feet from the table;
5. Lay the mast on its support in the center of the drill;
6. Make sure that all the control arms (Joysticks) are in the neutral position;
7. Apply the lockout procedure;
8. Remove the hydraulic hoses and install the numbered caps;
9. In order to balance the lifting point, four slings of equal length certified 5,000 pounds are attached to the four corners of the mast, at the appropriate places (rings). They must be linked together through the center with a large shackle one inch in diameter. Then approach the lifting equipment to attach the shackle;
10. Remove slack from slings without tensioning.
11. Proceed to remove the 12 actuator bolts **as follows until only 3 bolts remain:**



12. Recheck the tension on the slings;
13. Make sure you are not in the line of movement of the mast when it is raised
14. Remove the last three bolts;
15. Stand in a safe place at a minimum distance of twelve feet from the drill and make sure to be seen by the operator of the lifting equipment at all times;

VERY IMPORTANT
NEVER STAND UNDER THE MAST.
ALWAYS BE AWARE OF THE LINE OF FIRE OF ANY LIFTED PIECES

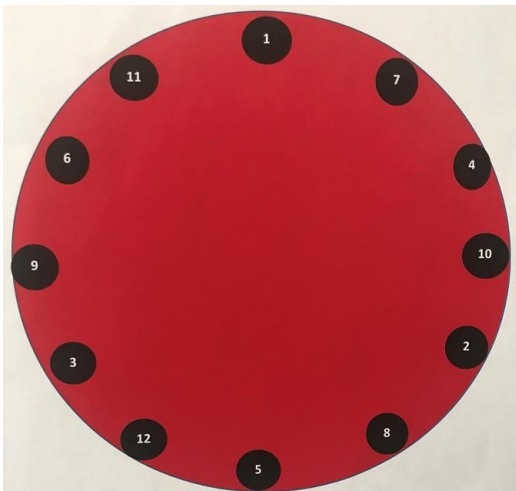
16. Remove the mast from the drill and place it on the ground on wooden blocks;
17. No one should remove the shackle until the mast is firmly seated on the blocking, then the shackle can be detached from the lifting equipment;
18. Retract the dump cylinders to the maximum
19. Disconnect the hydraulic hoses on the cylinders
20. Put caps on hoses and cylinders
21. Install a nylon sling as a "CHOKER" around the actuator plate
22. Remove the slack from the sling without tensioning;

WORK PROCEDURE FOR CHANGE, MAINTENANCE OR REPARATION ON THE CUBEX ACTUATOR

23. Unpin the cylinders at the 2 actuator ears;
24. Remove the two half-moons
25. Place the actuator on the ground, in a safe place to do the work

**BE AWARE OF YOUR FINGERS IN THE PINCHING POINTS
ALWAYS BE AWARE OF THE LINE OF FIRE OF ANY LIFTED PIECES**

26. Lift the actuator using the “CHOKE” sling
27. Align the top of the cylinders in their location and the support rods on the bottom of the half-moons
28. Replace the pins of the cylinders
29. Put the top of the half-moons back
30. Reconnect the hydraulic hoses
31. Reposition the actuator horizontally
32. “Time” the actuator
33. Insert the 12 bolts without tightening them
34. Tighten **the first 3 bolts in the following sequence**



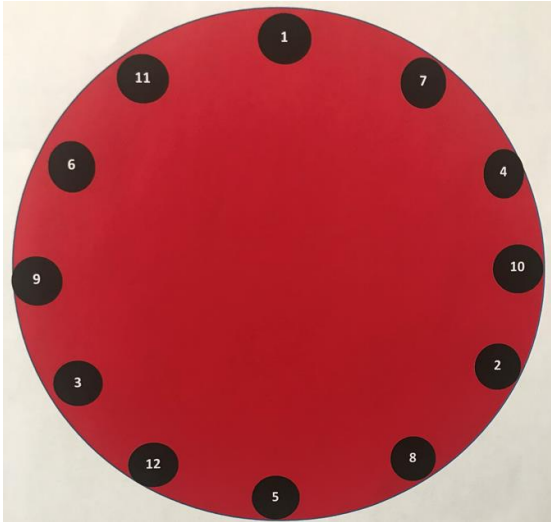
35. Then tighten the other 9 bolts according to the above sequence

**PLEASE NOTE THE BOLTS HAVE TO BE TIGHTENED AFTER 1 DRILLING SHIFT
APPLY A BLUE MECHANICAL TRACKING TAG TO THE DRILL CONTROLS**

PROCEDURE FOR CHANGING AN ACTUATOR IN A STOPE

1. Position the drill in a place where the floor is flat, the drill is stable and where it is possible to drill a very short hole;
2. Apply the stinger;
3. Drill a hole at least the length of the hammer in the rock
4. The hammer must remain in the hole with the gearbox as close to the tables as possible and the mast well supported on wood on the ground
5. Apply the lockout procedure;
6. Proceed to remove the 12 actuator bolts as follows until only 3 bolts remain:

WORK PROCEDURE FOR CHANGE, MAINTENANCE OR REPARATION ON THE CUBEX ACTUATOR



7. Make sure the mast is well anchored;
8. Remove the last three bolts;
9. Unlock
10. Turn the drill in the desired alignment so that the lifting equipment can approach
(You must carefully back the drill to separate the actuator from the mast, ensuring that there is no tension on the feed extension)
11. Move the "slide over" so that the lifting equipment is able to approach
12. Install a nylon sling as a "CHOKE" around the actuator plate;
13. Insert the lift fork in the free ring of the sling
14. Lift slowly to remove slack from the sling without putting tension;
15. Make sure the sling lifts vertically
16. Unpin the cylinders at the 2 actuator ears;
17. Remove the two half-moons;
18. Slowly lift the actuator;
19. Place the actuator on the ground in a safe place to do the work;

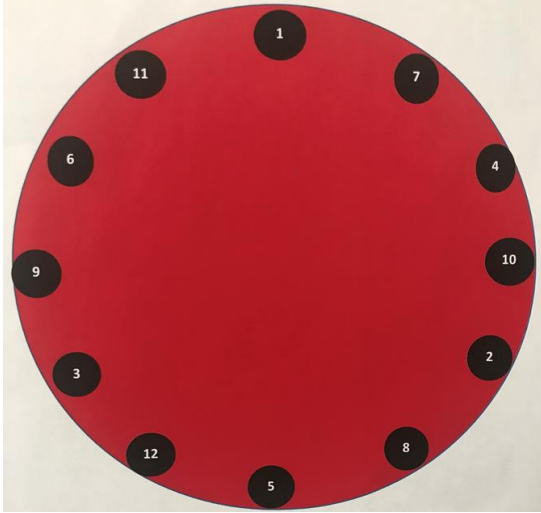
BE AWARE OF YOUR FINGERS IN THE PINCHING POINTS
ALWAYS BE AWARE OF THE LINE OF FIRE OF ANY LIFTED PIECES

To replace the actuator :

20. Lift the actuator using the "CHOKER" sling
21. Replace the pins of the dump cylinders
22. Put the top of the half-moons back
23. Reconnect the hydraulic hoses
24. Reposition the actuator horizontally
25. "Time" the actuator

WORK PROCEDURE FOR CHANGE, MAINTENANCE OR REPARATION ON THE CUBEX ACTUATOR

26. Insert the 12 bolts without tightening them
27. Tighten 3 bolts in the following sequence

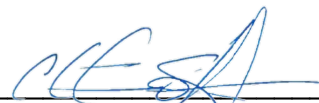


20. Then tighten the other 9 bolts according to the above sequence

**PLEASE NOTE THE BOLTS HAVE TO BE TIGHTENED AFTER 1 DRILLING SHIFT
APPLY A BLUE MECHANICAL TRACKING TAG TO THE DRILL CONTROLS**

Adds or modifications	Date	By
Translation review	May 2024	S. Tremblay R, Harrower


Karl Béchard
Directeur entretien


Christian St-Amour
Directeur des opérations